A Data-Driven Approach
to Financial Risk Management

Predicting Credit Card Defaults

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Introduction

The credit card industry plays a vital role in modern finance, facilitating flexible transactions.However, it faces significant risks from payment defaults, leading to major financial losses.

Crisis Overview (2006) in Taiwan: ReferenceLink
Aggressive lending practices and unchecked credit issuance led to a national credit card debt crisis.
Over 500,000 individuals were trapped in unmanageable debt cycles, causing societal issues like homelessness and increased suicide rates.

Objective

Proactively Manage Default Risks

Implement predictive analytics to identify high-risk customers early, enabling timely interventions

Enhance Portfolio Profitability

Reduce financial losses by focusing resources on managing default risks while targeting low-risk customers for promotional opportunities

Support Sustainable Credit Practices

Ensure fair lending policies and maintain customer trust by developing transparent, data-driven risk assessment models

Data preprocessing

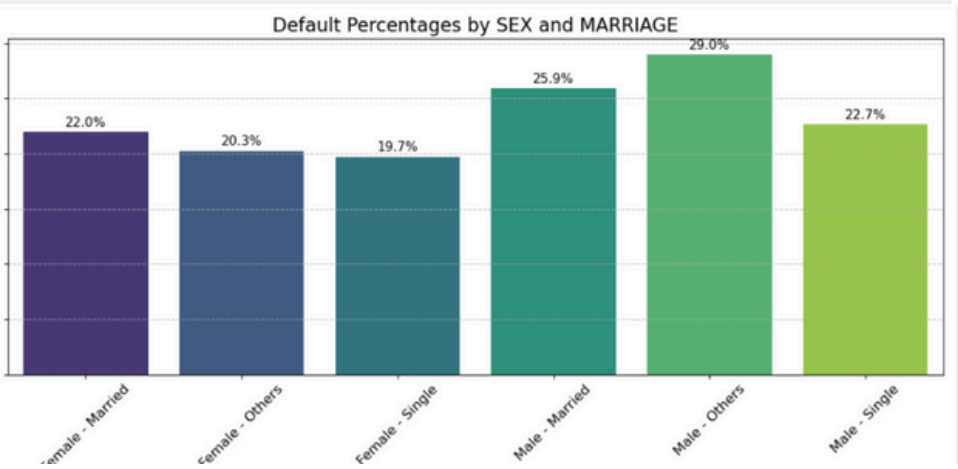
01 Merging very small groups in categorical variables, such as in the education and marriage columns

02 Mapping numerical values in repayment status, education, and marriage columns to descriptive categories

03 One-hot encoding to categorical variables like gender, education, marital status, and repayment statuses

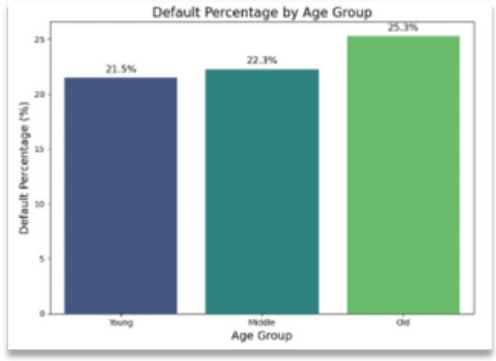
04 Data normalization for numerical columns

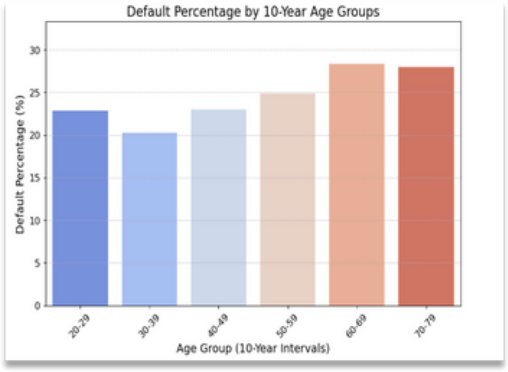
GENDER and MARITAL STATUS



- Male - Others :the highest default rate (29.0%), though they are a very small group.
- Married Males have the highest default rate among common groups (25.9%).
- Single Females have the lowest default rate (19.7%), indicating greater financial stability.

Age Group

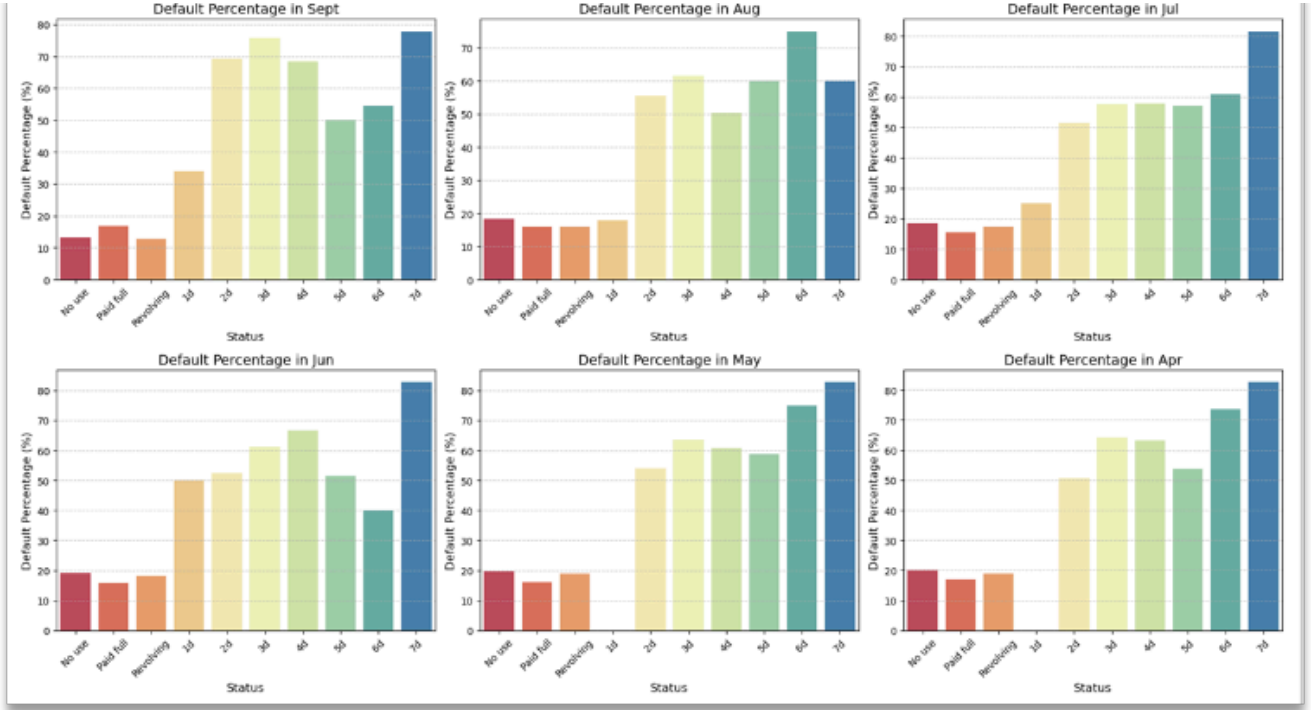




The default percentages showed increased risk with age,especially for the Older group.

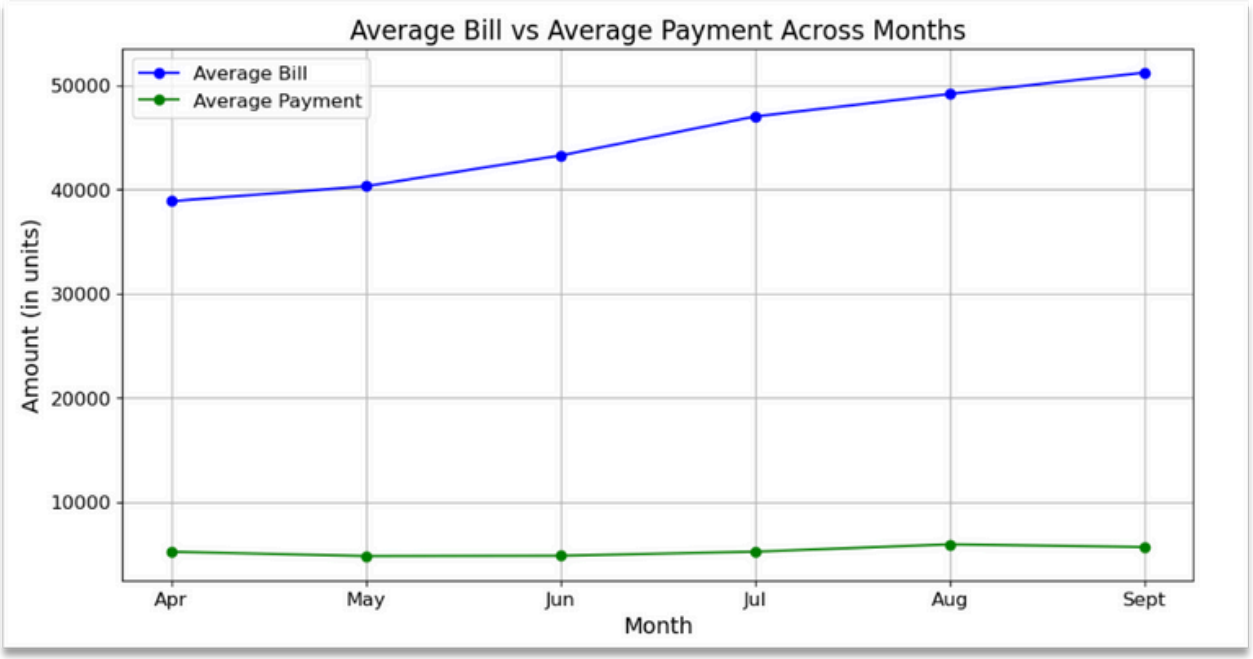
- Younger(20–29): Lower default rate (~23%)
- Middle-aged(50–59): Moderate default rate (~25%)
- Older (60–79): Highest default rate (~29%)

Repayment Status



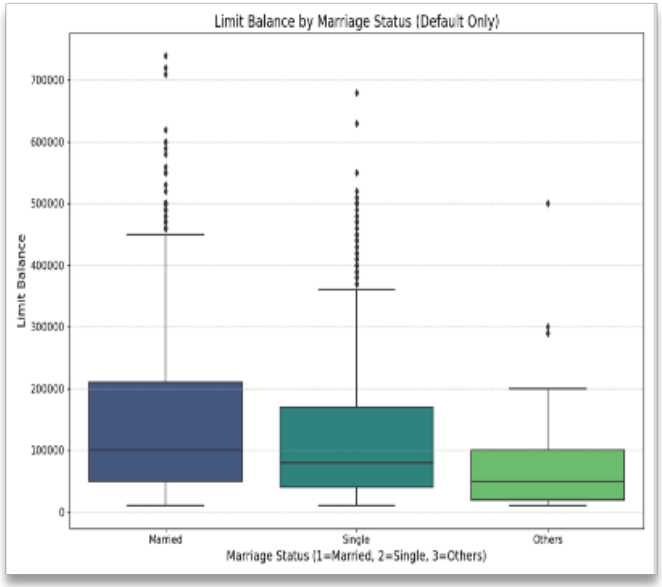
- Overall Trend: Severe delays (e.g., "7d") remain stable, with no significant rise or drop month-to-month.
- "7d" (7 months delayed): Consistently highest default percentage across months.
- "No use" & "Paid full": Lowest default percentages with minimal month-to-month variation.
- "1d" to "4d" (1–4 months delayed): Moderate defaults, showing slight variations but no drastic shifts.

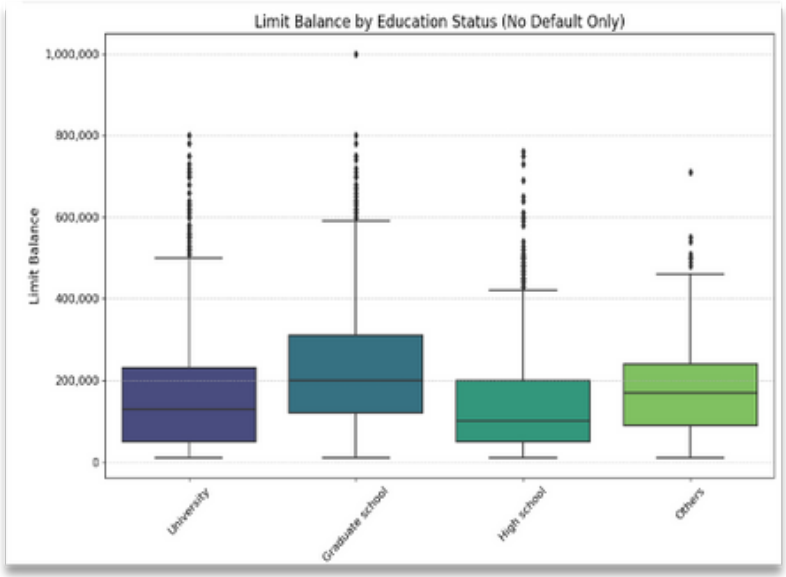
Average bill vs Average Payment



- Average payments stayed consistent, while bills steadily increased.
- This reinforces the observation that many could only afford minimum payments, regardless of their outstanding balances during Taiwan's credit card crisis

LIMIT BALANCE





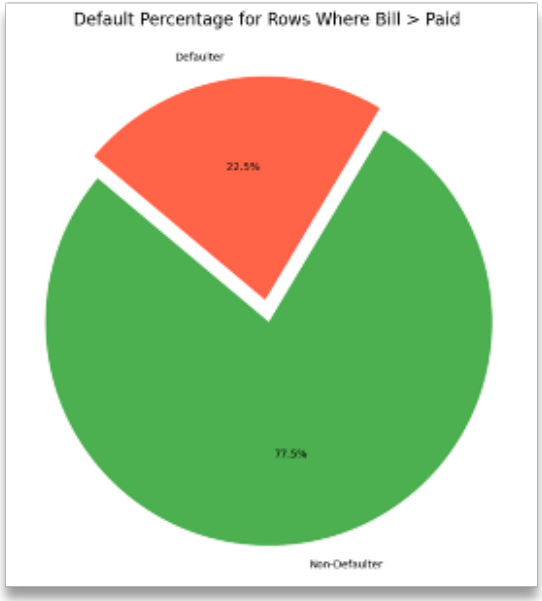
Marital Status: Challenges in Repayment

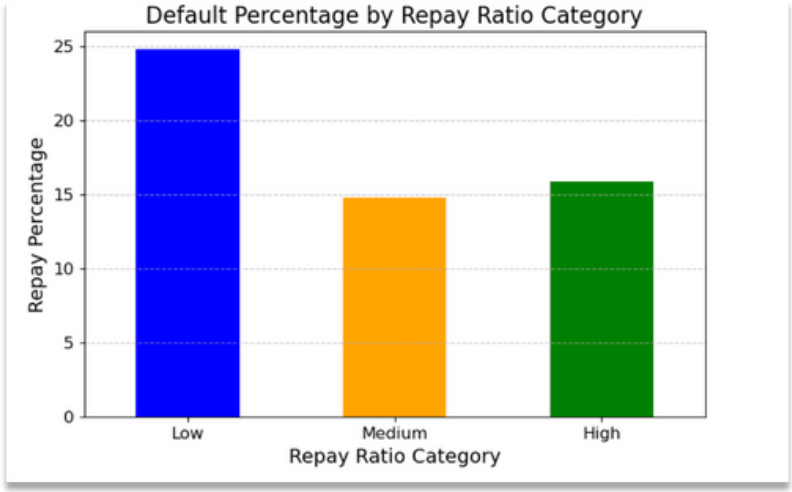
- Married individuals have higher credit limits, reflecting higher bills and family expenses, which may lead to repayment difficulties compared to singles.

Education Level: Insights into Financial Stability

- Graduate and University education levels are associated with higher credit limits, indicating greater earning potential and financial stability.
- High school education shows the lowest credit limits, suggesting reduced financial trust or capacity.

BILL > PAID





- 77.5% non-default rate despite bills exceeding payments.
- Default might be defined by prolonged non-repayment or high overdue balances.

- Individuals with lower repayment ratios are at a higher risk of default likely due to limited financial capacity or reliance on minimum payments.
- Higher repayment ratios are associated with lower default rates indicating better financial management and repayment behavior.

Logit Model (Cutoff:0.3)

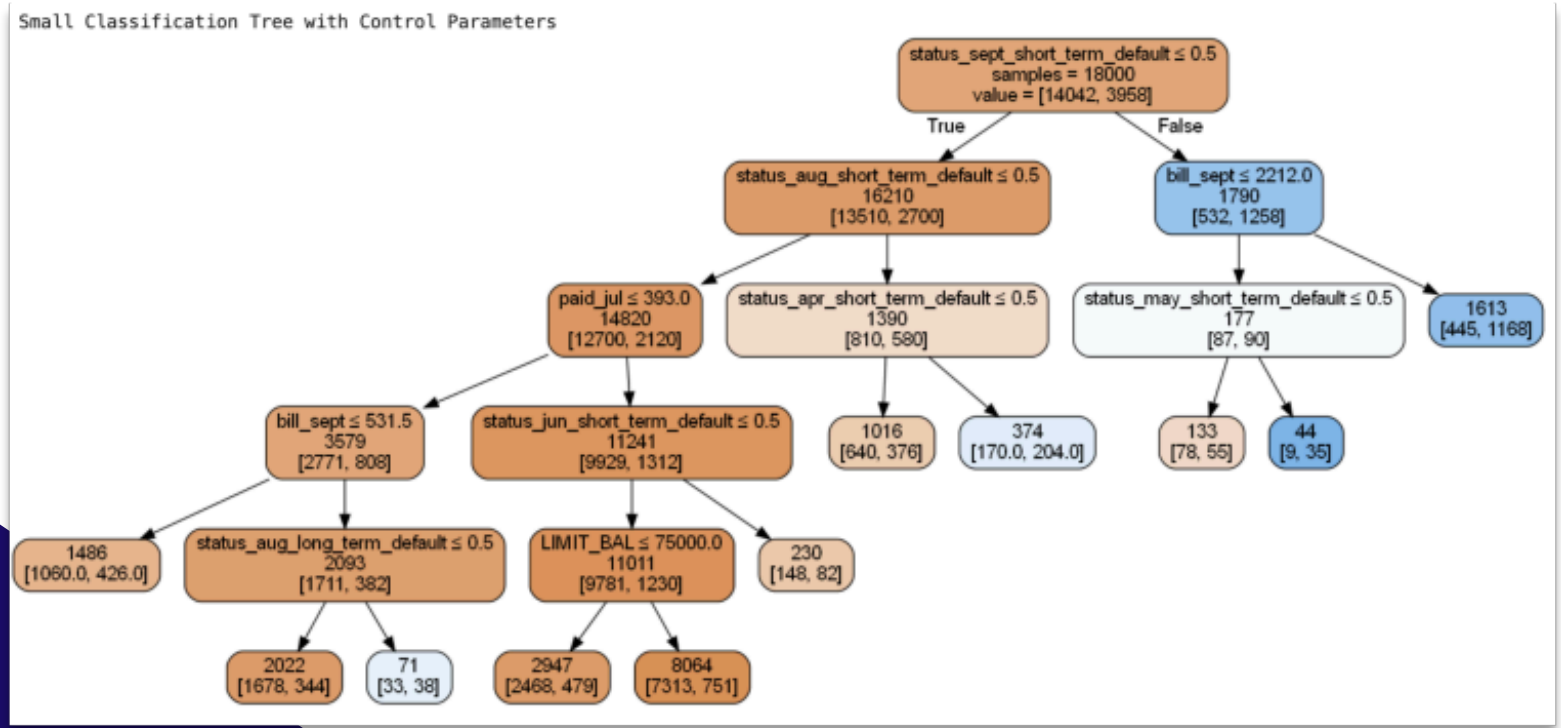
Logistic regression is ideal for this case as it provides clear probabilities for binary outcomes like default prediction, enabling effective risk management and decision-making.

Why Sensitivity Matters:
Sensitivity (48.2%) ensures defaults (true positives) are accurately identified.
False negatives (missed defaults) pose significant financial risks.
Trade-off with Specificity:
Specificity reduces to 89.4% is a trade off but considering the increase in sensitivity .
False positives are less harmful than missed defaults.

- Context:
- Taiwan's lenient credit approvals targeted young, low-income clients.
- This increased default risks, making early detection critical.
- Conclusion:
- Prioritizing sensitivity aligns with financial risk management.
- Logit Model 3 minimizes financial losses by identifying defaulters early.

Decision Tree Insights: Payment Delays and Defaults

- Short-Term Delays (2-3 months):
  - Significantly increase the chances of defaulting in the next month.
  - Represented by branches moving towards the right in the tree.
- First-Time Delays:
  - Result in considerably lower chances of default.
  - Represented by branches on the left of the tree.
- Conclusion:
  - Addressing short-term delays early can prevent defaults.



CREDIT HISTORY INDEX

CHI (Credit History Index) is a dynamic scoring system (0–120 points) that evaluates:

- Repayment behavior
- Credit utilization
- Payment-to-bill ratios

Why It Helps Credit Card Companies:

Categorizes Customer Base:

High-Risk: Likely to default (requires intervention).

Moderate-Risk: Manageable risk (guidance needed).

Low-Risk: Reliable customers (engagement opportunities).

Improves Risk Management:

Early identification of potential defaulters for proactive action.

Enhances Profitability:

Rewards reliable customers to boost credit usage and loyalty.

CHI Model A - Overview

- Starting CHI Score: 70 points.
- Adjustments:
- Payment Behavior: Penalize late payments and reward on-time payments.
- Credit Utilization:
  - 30–70%: +5 points (Moderate usage).
  - 70%: -8 points (High usage).
- Payment-to-Bill Ratio:
  - ≥90%: +6 points (Full payment).
  - <50%: -6 points (Low repayment).
- Output: CHI Score (0–120), categorized as:
- High-Risk (CHI < 20): Highest default risk.
- Moderate-Risk (CHI 20–60): Manageable risk.
- Low-Risk (CHI > 60): Most reliable clients.

Recommendation

01 Enhanced Application Screening: Strengthen approval processes with rigorous income verification, job stability criteria, and multi-tier income thresholds for credit limits.

02 Early Intervention Programs: Send automated reminders (calls, emails, SMS) before payment due dates to reduce missed payments.

03 Flexible Repayment Solutions: Offer structured repayment plans and penalty waiver programs to help clients manage overdue balances.

04 Proactive Credit Limit Management: Temporarily reduce credit limits for high-risk clients and flag accounts for close monitoring to detect repayment issues early.

Thank You